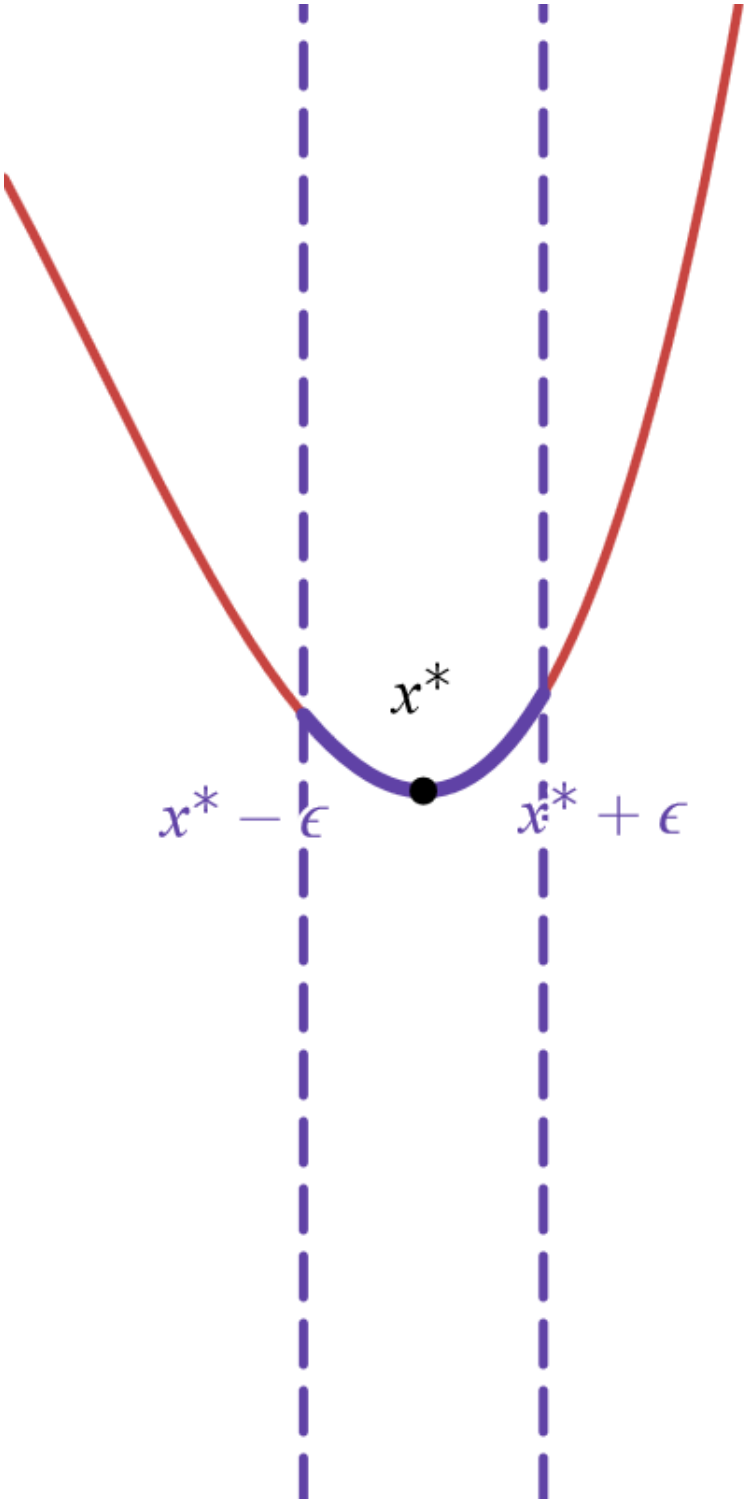
Lecture 6 (More Convexity, PL Functions, and Convex Sets)

January 29, 2026

First, we will prove a very useful property of convex functions.

Optimality in Convex Functions

Let $f : \mathbb{R}^n \to \mathbb{R}$. We say $\vec{x}^* \in \mathbb{R}^n$ is a **local minimum** if there exists $\epsilon > 0$ such that for all $\vec{x} \in \mathbb{R}^n$ such that $\|\vec{x} - \vec{x}^*\| < \epsilon$ we have $f(\vec{x}^*) \leq f(\vec{x})$.



This definition generalizes the calculus definition ($f'(x^*) = 0$ and $f''(x^*) > 0$), as it still allows for functions that are not differentiable everywhere. Also note that if $\vec{x}^*$ is a local minimum, this does not mean it is unique, that f is continuous, or a global minimum in general.

Theorem

If $f : \mathbb{R}^n \to \mathbb{R}$ is convex with a local minimum $\vec{x}^*$, then $\vec{x}^*$ must also be a global minimum.

Proof

Suppose $\vec{x}^*$ is a local minimum and $\vec{x}_0 \in \mathbb{R}^n$, such that $f(\vec{x}_0) < f(\vec{x}^*)$. Then as f is convex, we know for all $\lambda \in [0, 1]$ the following holds:

$$f\left(\lambda\vec{x}^* + (1-\lambda)\,\vec{x}_0\right) \leq \lambda f\left(\vec{x}^*\right) + (1-\lambda)\,f\left(\vec{x}_0\right) < \lambda f\left(\vec{x}^*\right) + (1-\lambda)\,f\left(\vec{x}^*\right) = f\left(\vec{x}^*\right)$$

Let $\vec{x} = \lambda\vec{x}^* + (1-\lambda)\,\vec{x}_0$. Then

$$\left\|\vec{x}^* - \vec{x}\right\| = \left\|\vec{x}^* - \left(\lambda\vec{x}^* + (1-\lambda)\,\vec{x}_0\right)\right\| = \left\|(1-\lambda)\,\vec{x}^* - (1-\lambda)\,\vec{x}_0\right\| = (1-\lambda)\left\|\vec{x}^* - \vec{x}_0\right\|$$

For any $\epsilon > 0$, we set $\max\left\{0, 1 - \frac{\epsilon}{\left\|\vec{x}^* - \vec{x}_0\right\|}\right\} < \lambda < 1$, thus

$$(1-\lambda)\left\|\vec{x}^* - \vec{x}_0\right\| < \left(1 - \left(1 - \frac{\epsilon}{\left\|x^* - \vec{x}_0\right\|}\right)\right)\left\|\vec{x}^* - \vec{x}_0\right\| = \frac{\epsilon}{\left\|x^* - \vec{x}_0\right\|}\left\|\vec{x}^* - \vec{x}_0\right\| = \epsilon$$

Thus

$$\left\|\vec{x}^* - \vec{x}\right\| < \epsilon$$

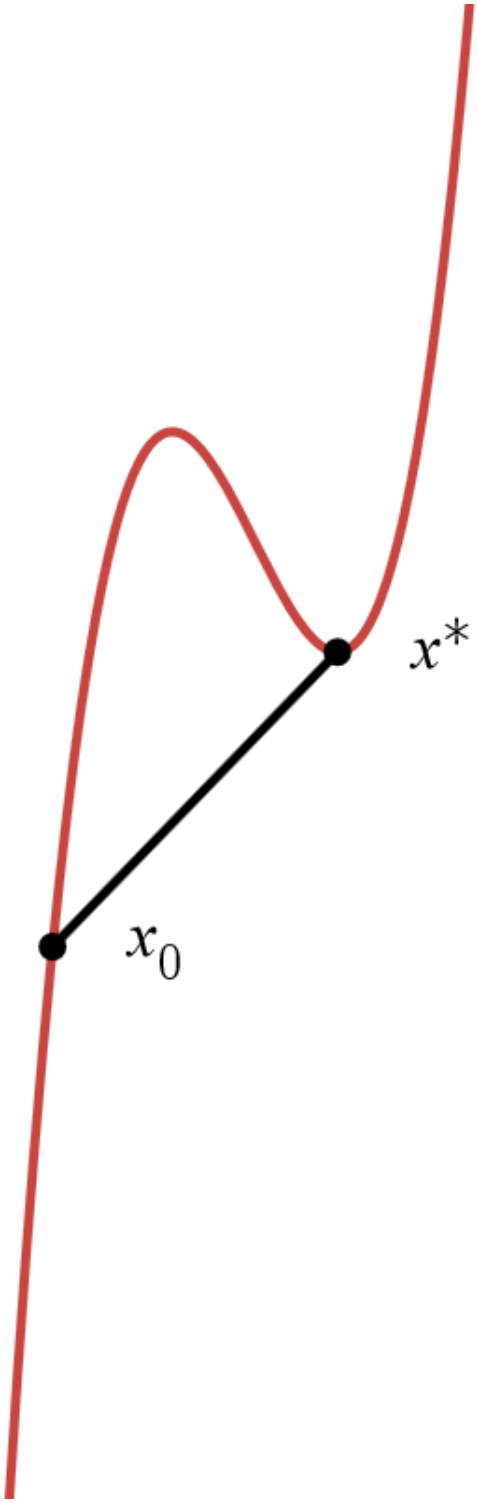
But we see

$$\vec{x} = \lambda\vec{x}^* + (1-\lambda)\,\vec{x}_0$$

with $\lambda \in [0, 1]$, thus

$$f\left(\vec{x}\right) = f\left(\lambda\vec{x}^* + (1-\lambda)\,\vec{x}_0\right) < f\left(\vec{x}^*\right)$$

This contradicts the assumption that $\vec{x}^*$ is a local minimum. To avoid contradiction, we then must assume $\vec{x}^*$ is a global minimum. We can see the contradiction visually below:



If $\vec{x}^*$ is a local minimum, we can see the line segment falls below the curve, so f cannot simultaneously have local minimum $\vec{x}^*$, be convex, and have $\vec{x}_0$ such that $f(\vec{x}_0) < f(\vec{x}^*)$. Also note that this does not prove the existence of a global minimum for convex f or that $\vec{x}^*$ is the unique global minimum. We only showed that if convex f has a local minimum then this local minimum must be a global minimum too.

PL Functions

A function $f : \mathbb{R}^n \to \mathbb{R}$ of the form

$$f(\vec{x}) = \max_{1 \leq i \leq n} \vec{a}_i \cdot \vec{x} + b_i$$

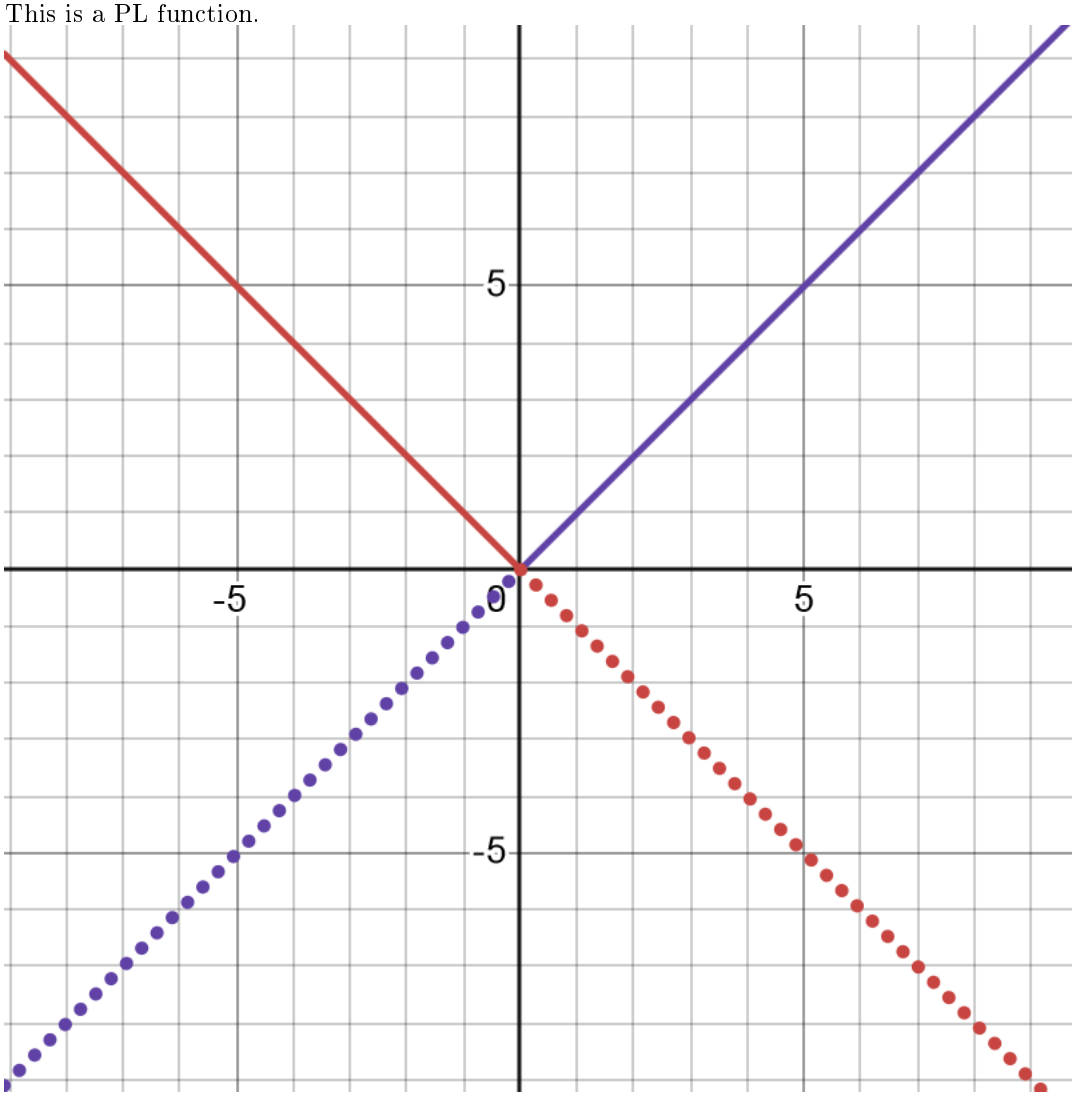
for some finite collection of $\vec{a}_i \in \mathbb{R}^n$ and $b_i \in \mathbb{R}$ is called a **piecewise linear convex function**. We usually abbreviate this to **PL Function**. In this class, if we say piecewise linear, we specifically mean a PL function unless specified otherwise.

Example

Consider

$$f(x) = |x| = \max\{x, -x\}$$

This is a PL function.



Non-Example

If we instead have $f(x) = -|x|$, this is no longer a PL function. In fact, we do not want to have any absolute values with negative coefficients in the objective function if the objective is a minimize objective. We cannot use the absolute value trick (discussed in previous lectures), and we will not be able to make any easy claims about minima, especially when combined with multiple absolute value functions and linear constraints.

Convex Sets

A set $S \subseteq \mathbb{R}^n$ is said to be **convex** if $\forall \vec{x}, \vec{y} \in S$ and for all $\lambda \in [0, 1]$ we have

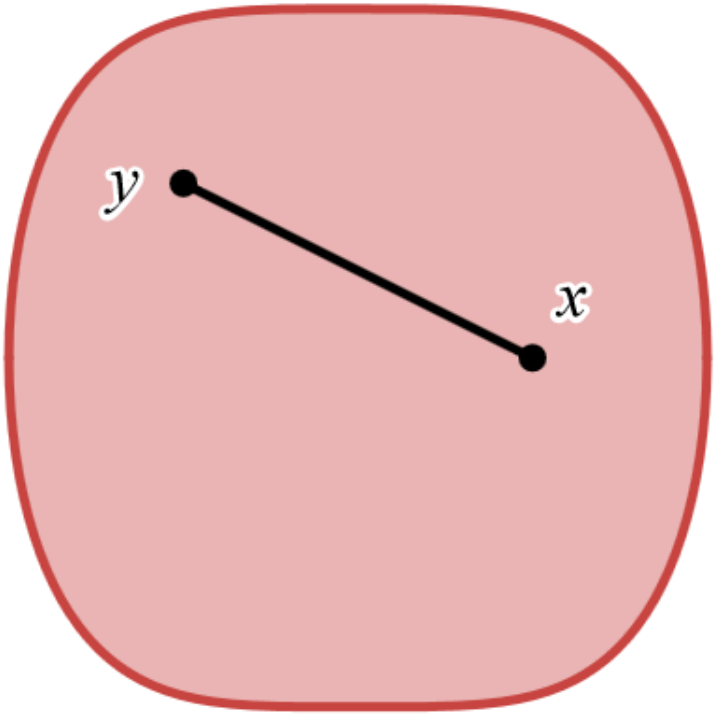
$$\lambda \vec{x} + (1 - \lambda)\vec{y} \in S$$

Geometrically, we can think of this as the property where the line segment joining any two vectors in S must lie entirely inside of S.

Examples

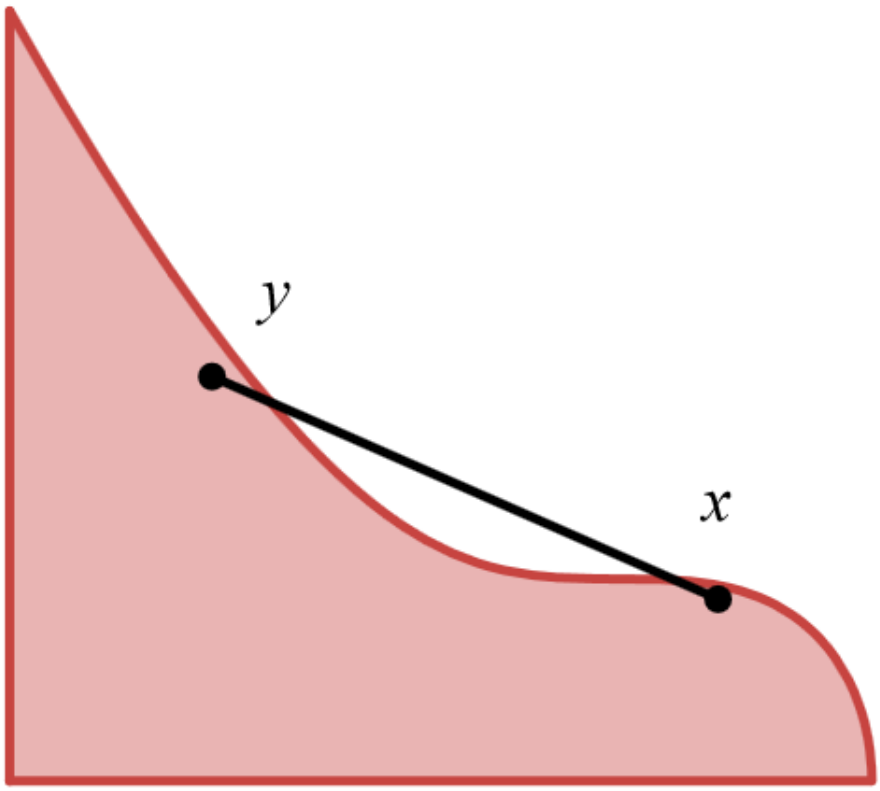
Ex 1)

Consider the following shape:



Notice any line segment joining two points in the shape must lie entirely within the shape.

Non Ex 2)



Notice that the line segment joining x and y crosses outside of the shape. This means this shape is not convex.

Ex 3)

We can also talk about convexity of sets more abstractly. Let

$$S = \left\{ \vec{x} \in \mathbb{R}^n | \vec{x} > \vec{b} \right\}$$

S is convex. To show this, Let $\vec{x}, \vec{y} \in S$. Also let $\lambda \in [0, 1]$. Then

$$\lambda\vec{x} + (1 - \lambda)\,\vec{y} > \lambda\vec{b} + (1 - \lambda)\,\vec{b} = \vec{b}$$

as $\lambda, (1-\lambda) \geq 0$.

Ex 4)

We can even talk about matrices! Let

$$S = \left\{ A \in \mathbb{R}^{n \times n} | \sum_i \sum_j A_{ij} \leq b \right\}$$

S is convex. Let $A, B \in S$. Let $\lambda \in [0,1]$. Let

$$C = \lambda A + (1-\lambda) B$$

Then

$$C_{ij} = \lambda A_{ij} + (1-\lambda) B_{ij}$$

Thus

$$\sum_i \sum_j C_{ij} = \sum_i \sum_j \lambda A_{ij} + (1-\lambda) B_{ij} = \lambda \sum_i \sum_j A_{ij} + (1-\lambda) \sum_i \sum_j B_{ij} \leq \lambda b + (1-\lambda) b = b$$

Thus

$$C \in S$$

Some Important Convex Sets

Now we will talk about some convex sets that are very important for our purposes. A **polyhedron** is a set S that can be described in the form:

$$S = \left\{ \vec{x} \in \mathbb{R}^n | A\vec{x} \geq \vec{b} \right\}$$

for some $A \in \mathbb{R}^{m \times n}$ and some $\vec{b} \in \mathbb{R}^m$. We can equivalently define a polyhedron as a set that can be written in the form:

$$S = \left\{ \vec{x} \in \mathbb{R}^n | A\vec{x} \leq \vec{b} \right\}$$

or a set that can be written in the form:

$$S = \left\{ \vec{x} \in \mathbb{R}^n | A\vec{x} = \vec{b}, \vec{x} \geq 0 \right\}$$

The last form is called **standard form**. Note how similar these sets look to our linear constraints for LPs. Also note that polyhedra are allowed to be infinite. We say a polyhedron, S, is **bounded** if there exists a $K \in \mathbb{R}$ such that

$$\|\vec{x}\| > k \ \forall \vec{x} \in S$$

If a polyhedron is not bounded, we say it is **unbounded**. We also have a couple of special cases of polyhedra.
a) A set that can be written in the form

$$S = \{ \vec{x} \in \mathbb{R}^n | \vec{a} \cdot \vec{x} = b \}$$

for some $\vec{a} \in \mathbb{R}^n$, $b \in \mathbb{R}$ is called a **hyperplane**.
b) A set that can be written in the form

$$S = \{ \vec{x} \in \mathbb{R}^n | \vec{a} \cdot \vec{x} \geq b \}$$

for some $\vec{a} \in \mathbb{R}^n$, $b \in \mathbb{R}$ is called a **half plane** or a **half space**. Consider a finite collection of vectors in $\mathbb{R}^n$:

$$X = \{ \vec{x}_1, \vec{x}_2, \ldots, \vec{x}_k \}$$

Also consider a collection $\lambda_i \in [0,1]$ such that $\sum_{i=1}^k \lambda_i = 1$. Then we call

$$\sum_{i=1}^k \lambda_i \vec{x}_i$$

a **convex combination** of vectors in X. If we take all possible convex combinations, we get the **convex hull** of X

$$S = \left\{ \vec{y} | \vec{y} = \sum_{i=1}^k \lambda_i \vec{x}_i, \ \vec{x}_i \in X, \ \sum_{i=1}^k \lambda_i = 1, \ 0 \leq \lambda_i \leq 1 \right\}$$

Theorem

Now let's prove some important properties of convex sets.

a) The intersection of convex sets is convex.
b) Every polyhedron is a convex set
c) A convex combination of a finite number of elements in a convex set, S, belongs in S
d) The convex hull of a finite collection of vectors is a convex set.

Proof

a)

Let I be an indexing set and let S_i be convex for all $i \in I$. Consider

$$S = \cap_{i \in I} S_i$$

Suppose

$$x, y \in S,\ \lambda \in [0,1]$$

Then

$$x, y \in S \implies x, y \in S_i\ \forall i \in I$$

Thus

$$\lambda x + (1-\lambda)\, y \in S_i\ \forall i \in I$$

$$\lambda x + (1-\lambda)\, y \in \cap_{i \in I} S_i = S$$

Thus S is convex.

b)

Let $S = \left\{\vec{x} | A\vec{x} \geq \vec{b}\right\}$. Let $\vec{x}, \vec{y} \in S$ and $\lambda \in [0,1]$. Then

$$A\left(\lambda\vec{x} + (1-\lambda)\,\vec{y}\right) = \lambda A\vec{x} + (1-\lambda)\, A\vec{y} \geq \lambda\vec{b} + (1-\lambda)\,\vec{b} = \vec{b}$$

As $\lambda, (1-\lambda) \in [0,1]$

c)

We will prove this by induction on the number of elements in the convex combination. Let S be convex.

Base Case: $n = 2$

Suppose $\vec{x}_1, \vec{x}_2 \in S$, $\lambda_1, \lambda_2 \in [0,1]$, and $\lambda_1 + \lambda_2 = 1$. Then

$$1 - \lambda_1 = \lambda_2$$

Thus

$$\sum_{i=1}^{2} \lambda_i \vec{x}_i = \lambda_1\vec{x}_1 + \lambda_2\vec{x}_2 = \lambda_1\vec{x}_1 + (1-\lambda_1)\,\vec{x}_2 \in S$$

as S is convex.

Induction Step

Suppose $\sum_{i=1}^{m} \lambda_i\vec{x}_i$ holds for all $1 \leq m \leq n$, when $\sum_{i=1}^{m} \lambda_i = 1$, $\lambda_i \in [0,1]$, and $\vec{x}_i \in S$ for all $1 \leq i \leq m$. Now, consider

$$\sum_{i=1}^{n+1} \lambda_i\vec{x}_i$$

as a convex combination of $\vec{x}_i \in S$ for $1 \leq i \leq n+1$. Then

$$\sum_{i=1}^{n+1} \lambda_i\vec{x}_i = \lambda_{n+1}\vec{x}_{n+1} + \sum_{i=1}^{n} \lambda_i\vec{x}_i$$

If $\lambda_{n+1} = 1$, then

$$\sum_{i=1}^{n+1} \lambda_i = \lambda_{n+1} + \sum_{i=1}^{n} \lambda_i = 1 + \sum_{i=1}^{n} \lambda_i = 1$$

Thus

$$\sum_{i=1}^{n} \lambda_i = 0$$

And

$$\sum_{i=1}^{n+1} \lambda_i \vec{x}_i = \lambda_{n+1}\vec{x}_{n+1} + \sum_{i=1}^{n} \lambda_i \vec{x}_i = \lambda_{n+1}\vec{x}_{n+1} + 0 = \lambda_{n+1}\vec{x}_{n+1} \in S$$

trivially. Now, suppose $\lambda_{n+1} < 1$. Then

$$\sum_{i=1}^{n+1} \lambda_i = \lambda_{n+1} + \sum_{i=1}^{n} \lambda_i = 1$$

implies

$$\sum_{i=1}^{n} \lambda_i = 1 - \lambda_{n+1}$$

$$\sum_{i=1}^{n} \frac{\lambda_i}{(1-\lambda_{n+1})} = 1$$

Thus

$$\sum_{i=1}^{n+1} \lambda_i \vec{x}_i = \lambda_{n+1}\vec{x}_{n+1} + \sum_{i=1}^{n} \lambda_i \vec{x}_i = \lambda_{n+1}\vec{x}_{n+1} + (1-\lambda_{n+1}) \sum_{i=1}^{n} \frac{\lambda_i}{(1-\lambda_{n+1})} \vec{x}_i$$

and by the induction hypothesis:

$$\sum_{i=1}^{n} \frac{\lambda_i}{(1-\lambda_{n+1})} \vec{x}_i = \vec{x} \in S$$

Thus

$$\lambda_{n+1}\vec{x}_{n+1} + (1-\lambda_{n+1}) \sum_{i=1}^{n} \frac{\lambda_i}{(1-\lambda_{n+1})} \vec{x}_i = \lambda_{n+1}\vec{x}_{n+1} + (1-\lambda_{n+1})\,\vec{x} \in S$$

As $\vec{x}, \vec{x}_{n+1} \in S$ and S is convex.

d)

Left as homework exercise.